

Task 1 - Biometric Attendance System

Project Title

Biometric Attendance System using Fingerprint Sensor and Arduino

Objective

To design a biometric attendance system that records attendance using fingerprint authentication and stores logs with date/time.

Duration

6 Hours

Project Complexity

Easy

Learning Outcomes

- Understand biometric authentication.
- Learn sensor interfacing with Arduino.
- Real-time attendance logging.
- Serial/database data handling.

Components Required

- Arduino Uno
- Fingerprint Sensor (R307 / R305)
- LCD Display (optional)
- RTC Module (optional)
- Buzzer
- Jumper Wires
- USB Cable
- Computer with Arduino IDE
- Database (MySQL / Excel / CSV optional)

Working Principle

1. User places finger on sensor.
2. Sensor scans fingerprint.
3. Arduino compares with stored templates.

4. If matched, attendance marked.
5. Time and user ID logged.
6. Buzzer confirms success.

Circuit Connections

- Sensor TX -> Arduino D2
- Sensor RX -> Arduino D3
- VCC -> 5V
- GND -> GND
- Buzzer + -> D8
- Buzzer - -> GND

Arduino Code

```
#include <Adafruit_Fingerprint.h>
#include <SoftwareSerial.h>

SoftwareSerial mySerial(2, 3);
Adafruit_Fingerprint finger = Adafruit_Fingerprint(&mySerial);

int buzzer = 8;

void setup() {
  pinMode(buzzer, OUTPUT);
  Serial.begin(9600);
  finger.begin(57600);

  if (finger.verifyPassword()) {
    Serial.println("Fingerprint Sensor Found");
  } else {
    Serial.println("Sensor Not Found");
    while (1);
  }
}

void loop() {
  int id = getFingerprintID();

  if (id > 0) {
    Serial.print("Attendance Marked for ID: ");
    Serial.println(id);
    digitalWrite(buzzer, HIGH);
    delay(500);
    digitalWrite(buzzer, LOW);
    delay(2000);
  }
}
```

```

    }
}

int getFingerprintID() {
    uint8_t p = finger.getImage();
    if (p != FINGERPRINT_OK) return -1;

    p = finger.image2Tz();
    if (p != FINGERPRINT_OK) return -1;

    p = finger.fingerFastSearch();
    if (p != FINGERPRINT_OK) return -1;

    return finger.fingerID;
}

```

Sample Output

```

Fingerprint Sensor Found
Attendance Marked for ID: 1
Attendance Marked for ID: 3

```

Real-World Applications

- College attendance system
- Office employee tracking
- Secure room access control
- Lab entry monitoring

Future Enhancements

- Cloud database integration
- Face recognition using Raspberry Pi
- SMS notification
- Web dashboard reporting

Conclusion

This project provides secure and automated attendance management using biometric verification. It reduces proxy attendance and manual errors.